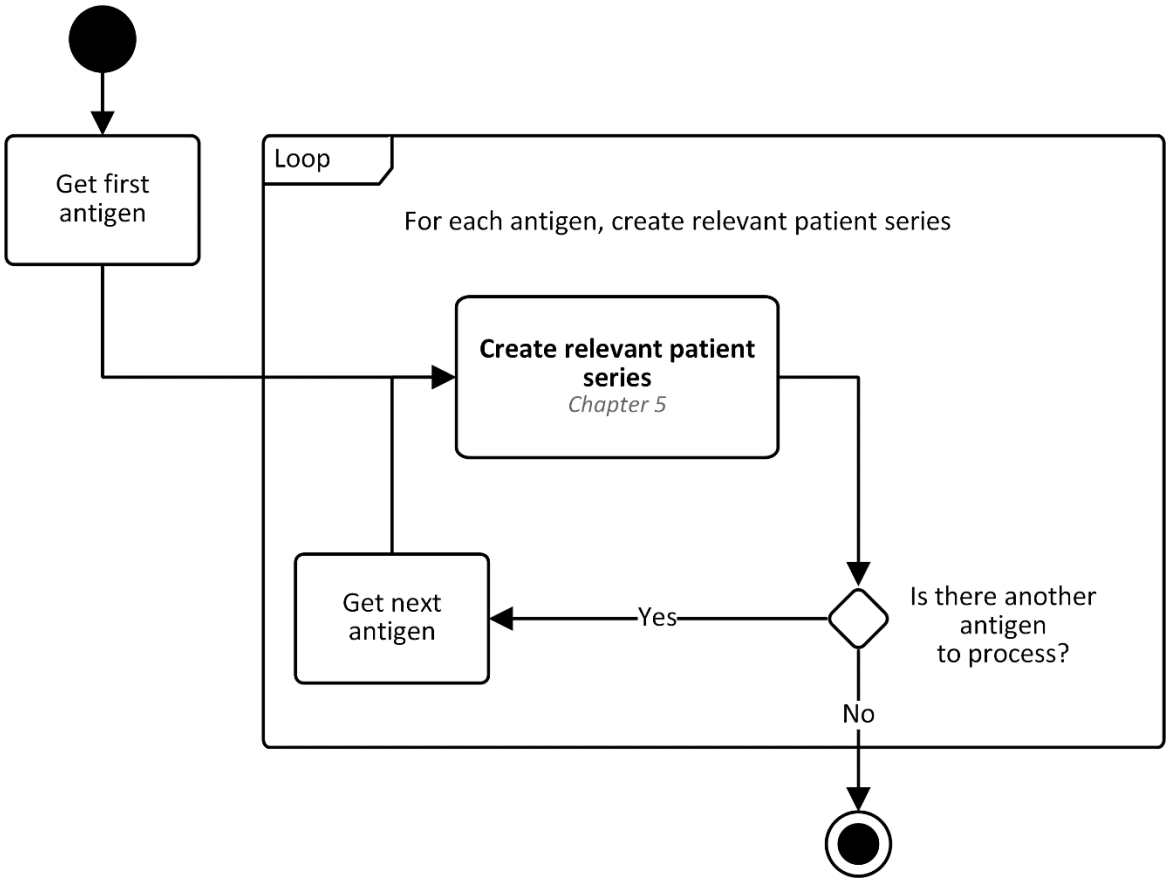

FIGURE 4-4 CREATE RELEVANT PATIENT SERIES PROCESS MODEL

4.4 Evaluate and Forecast all Relevant Patient Series

This step is the core of the business logic and decision points many people think of when describing evaluation and forecasting. In the Logic Specification, this step contains all of the clinical business rules and decision logic in the form of business rules and decision tables.

At the end of this step, each relevant patient series will have an evaluated history and a forecast.

The iterative nature of this step is best described with two activity diagrams. First, Figure 4-5 shows the high-level iterative process of looping through all relevant patient series. Next, Figure 4-6 specifically deals with the details of evaluation. A description of the activity diagram follows each figure.

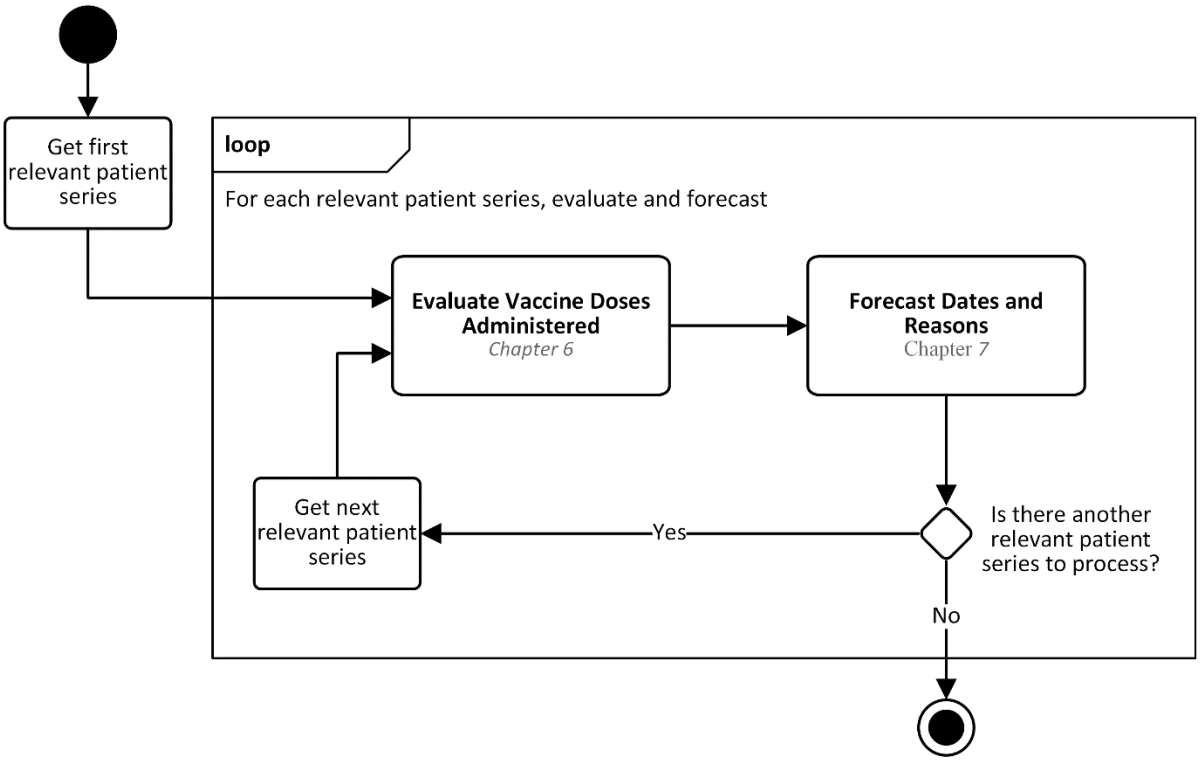

FIGURE 4-5 EVALUATE AND FORECAST PROCESS MODEL

At the highest level of this step, as illustrated in the figure above, a simple iterative process is used to walk through each relevant patient series and apply the logic defined in the evaluation and forecasting chapters.

For each relevant patient series created in the *create relevant patient series* step (Chapter 5), the following steps are performed:

1. Evaluate the immunization history. See the *evaluate immunization history* activity diagram below for further details.
2. Create forecast dates and/or reasons for the next target dose to be administered. Process models and detailed decision logic on forecasting are in Chapter 7.

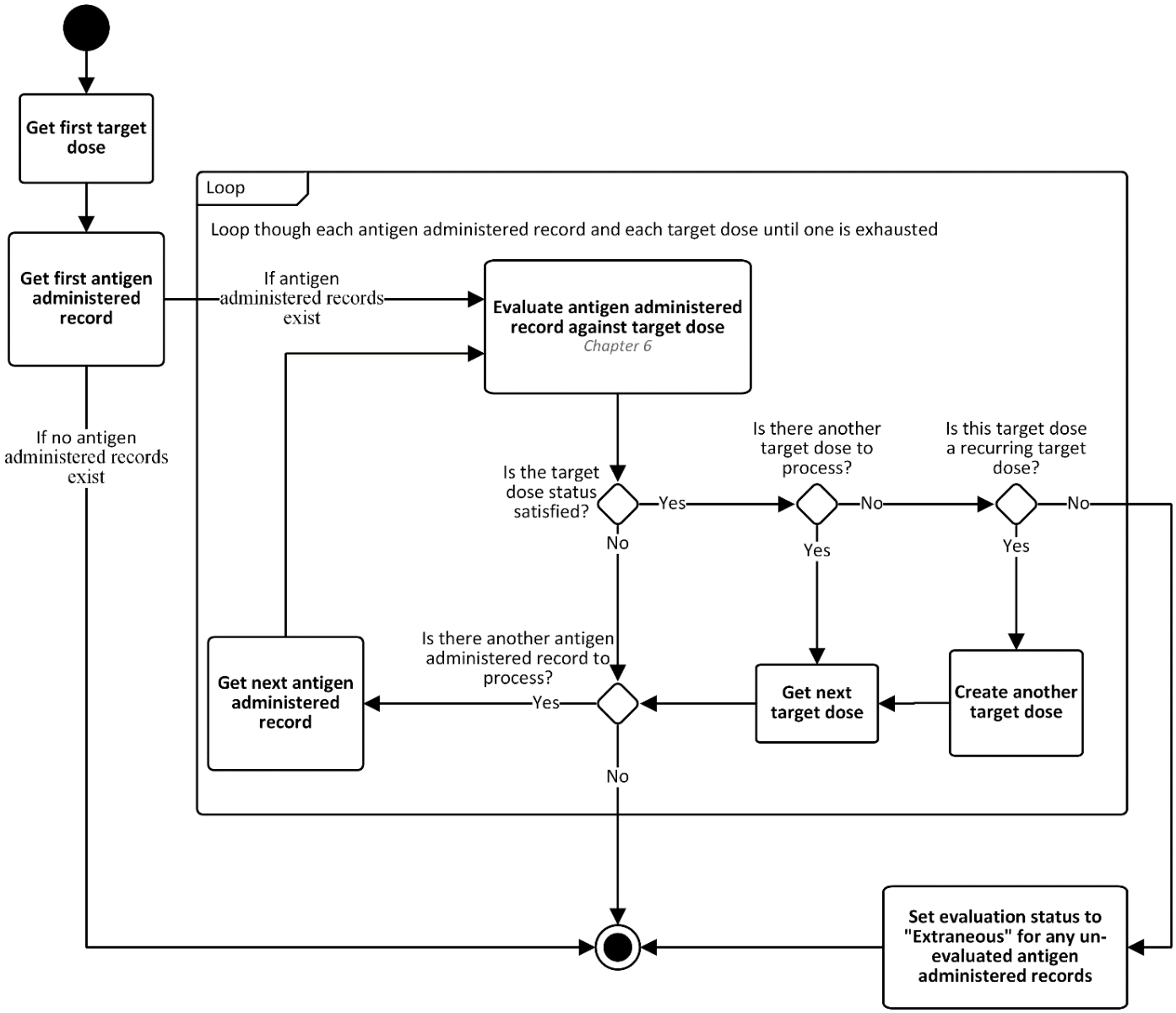

FIGURE 4-6 EVALUATE IMMUNIZATION HISTORY PROCESS MODEL

Figure 4-6 illustrates the iterative nature of *evaluate immunization history* in greater detail. There are two collections (arrays, lists, etc.) which must be traversed. The first collection is the relevant patient series consisting of one or more target doses. The second collection is the antigen administered records. At any point in the iterative process either collection could be the trigger to end our evaluation process. Specifically, whichever collection is exhausted first will be the trigger for ending the evaluation process.

It is important to note the contents of antigen administered records at this point in the process. Antigen administered records are only those which could potentially satisfy the goals of the relevant patient series. For example, if the patient series is a path to immunity for HepB, then the antigen administered records will only contain HepB records in ascending date order.

It should also be noted that when multiple relevant patient series have been created (Chapter 5), all antigen administered records for the patient should be evaluated against each relevant patient series, and a status (valid,

not valid, etc.) should be assigned per dose per relevant patient series. An administered dose that is “valid” for one relevant patient series may be “not valid” for a different relevant patient series for the same patient.

The evaluate immunization history process is as follows:

1. The process begins by getting the first target dose from the relevant patient series collection. The current target dose is an important concept as the process moves from evaluation into forecasting. The evaluation process will inform the forecasting process which target dose needs to be forecasted.
2. If the antigen administered collection has elements in it, the process gets the first antigen administered and continues to step 3.
    a. If the antigen administered collection is empty, the evaluation process for this relevant patient series ends.
3. The step described as “evaluate the antigen administered record against the target dose” is a reference to Chapter 6 which contains process models and detailed decision logic that must be followed prior to moving on to step 4.
4. After the antigen administered record was evaluated against the target dose, the next step is to determine which collections to iterate based on the results of the evaluation.
    a. If the target dose status is satisfied, proceed to step 5.
        i. The antigen administered was valid. The target dose is satisfied. The evaluation process can push forward to the next target dose if one exists.
    b. If the target dose status is not satisfied, proceed to step 7.
        i. The antigen administered did not meet the goals of the target dose. The evaluation process cannot move onto the next target dose.
5. This step determines if there are more target doses in the relevant patient series collection.
    a. If the relevant patient series collection has been exhausted, proceed to step 6.
    b. If the relevant patient series collection contains another target dose, get the next target dose and proceed to step 7.
6. This step determines if the current target dose (now the last target dose in the relevant patient series) is a recurring dose. This is a semi-rare condition (e.g., Td boosters, yearly Flu, certain risk series). A recurring dose may recur based on a time interval from the previous dose (i.e. a tetanus recurring dose every 10 years for adults) or based on a patient observation (i.e. a pertussis recurring dose with every pregnancy).
    a. If the target dose is defined to be a recurring dose, initialize a new target dose identical to the current target dose. The newly created target dose must now be the last element in the collection. Finally, iterate the collection to get this target dose and proceed to step 7.
    b. If the target dose is not defined to be a recurring dose, the evaluation process for this relevant patient series ends. Any remaining antigen administered records should have their evaluation statuses set to “extraneous.”
7. This step determines if there are any more antigen administered records to evaluate.
    a. If the antigen administered collection has been exhausted, the evaluation process for this relevant patient series ends.
    b. If the antigen administered collection contains another record, get the next antigen administered record and return back to step 3.
        i. Repeat steps 3 – 7 until the evaluation process for this relevant patient series ends. At this point the process can end in one of two ways: (1) No more target doses (step 6.b) or (2) No more antigen administered records (step 7a).